Engineered Biocomposites for Angiogenic Signaling and Tissue Vascularization

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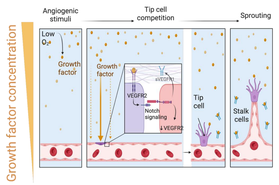
Background

The Problem

- Cells beyond 200 µm from the nearest capillary are susceptible to hypoxia and necrosis
- Vascularization remains the critical bottleneck in engineering viable 3D tissue constructs

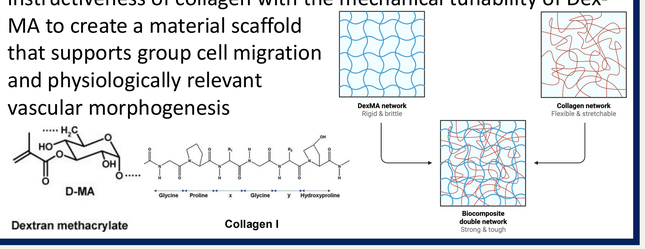
The Challenge

- Endothelial cells require coordinated biophysical and biochemical matrix cues to form vasculature
- Matrix stiffness must be precisely tuned – too rigid prevents migration; too soft disrupts paracrine signaling essential to collective cell behavior



Our Approach

- Collagen I/Dex-MA biocomposite hydrogels leverage the biological instructiveness of collagen with the mechanical tunability of Dex-MA to create a material scaffold that supports group cell migration and physiologically relevant vascular morphogenesis

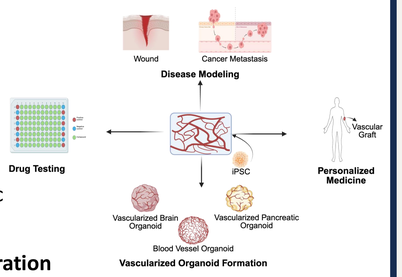


Objectives

1. **Characterize** how the biophysical properties of Collagen I and Dex-MA composites (e.g., stiffness, porosity, network architecture) influence HUVEC and HDF spreading and migration.
2. **Identify** hydrogel formulations that promote coordinated, group cell translocation and *in situ* vascular network formation in co-culture.
3. **Establish** design principles linking scaffold composition to vasculogenic outcome, laying groundwork for future paracrine signaling studies and organoid applications.

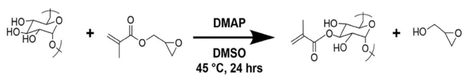
Applications

- Physiologically relevant **screening platform**
- Tunable matrix enables **disease-state mimicry**
- Scalable 3D tissue vascularization
- Individualized therapeutic response profiling
- Patient derived **cell integration**

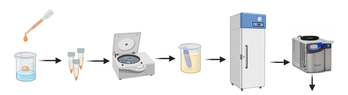


Methods

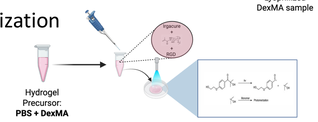
- DexMA synthesis



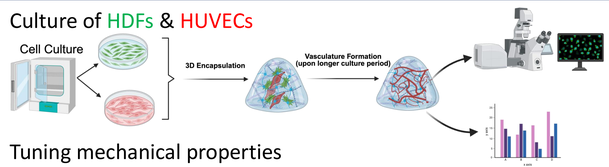
- Product purification



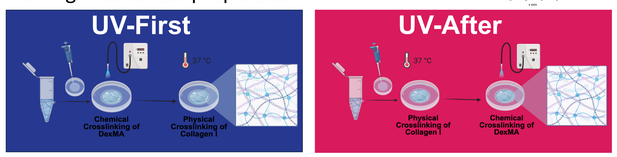
- Resolubilization



- Culture of HDFs & HUVECs



- Tuning mechanical properties

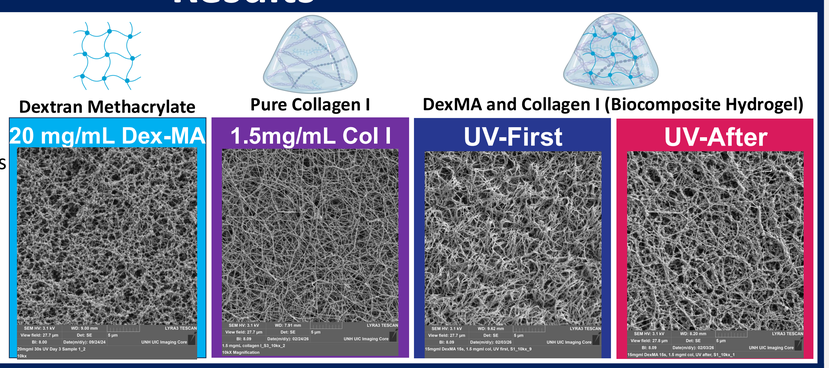


- Analysis using SEM, Rheometer, Confocal Microscopy

Results

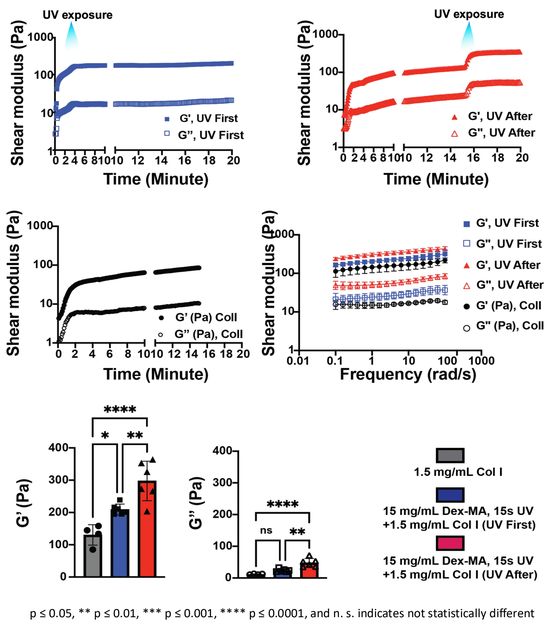
Gel Microstructures

- Defined structural differences between pure materials (Dex-MA/Collagen I) & biocomposites (UV-First/UV-After)
- Defined structural differences between UV-First and UV-After conditions



Tunable Stiffness

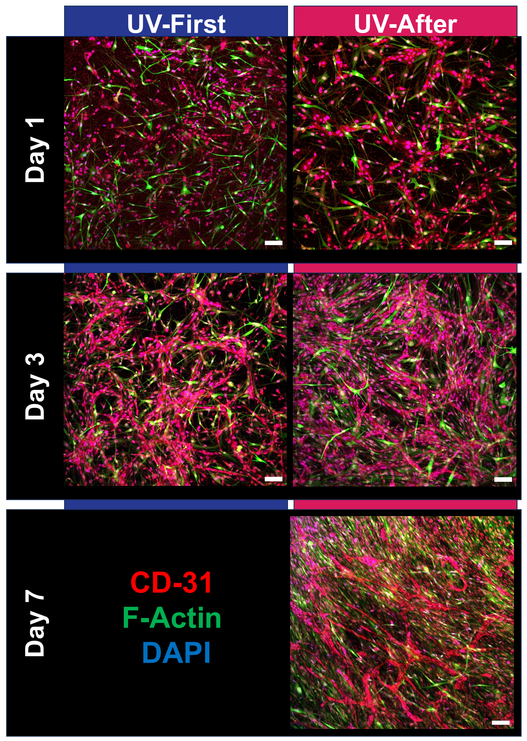
- Mechanical study performed on oscillatory rheometer
- Material properties recorded during *in situ* gelation



p ≤ 0.05, ** p ≤ 0.01, *** p ≤ 0.001, **** p ≤ 0.0001, and n. s. indicates not statistically different

In Situ Vasculature Formation

- Confocal microscopy used to image large-scale organization of HUVECs and HDFs

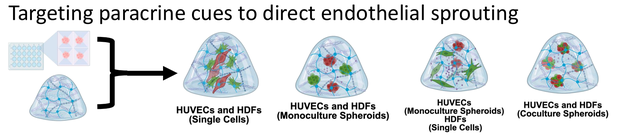


Discussion

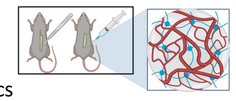
- Dex-MA & Collagen biocomposite hydrogel has **distinct mechanical properties** compared to Dex-MA only/Collagen only hydrogels
- UV-First promotes a **less interconnected** network (lower G')
- UV-After promotes a **more interconnected** network (higher G')
- Biocomposite hydrogel (Dex-MA & Collagen I) facilitates **capillary-like structure formation**
- **UV-First and Collagen only** hydrogels contract over a longer culture period and thus are **not suitable for vessel-like structure formation**
- **UV-After** (15 mg/mL Dex-MA, 15s UV, 1.5 mg/mL Collagen I) is **suitable** for HUVECs and HDFs single cell coculture for vessel-like structure formation
- **Coculture spheroids** in biocomposite hydrogels will **mimic the 3D** physiological conditions **more precisely**

Future Work

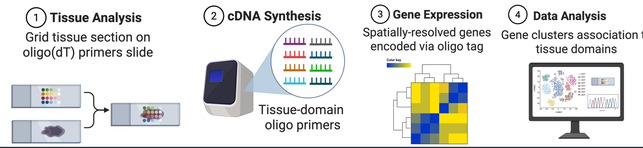
- Targeting paracrine cues to direct endothelial sprouting



- Quantitative and qualitative sprout analysis
- *In vivo* animal model



- Spatial Transcriptomics



Acknowledgements

Li Lab Group

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